

HYAKURI AB, Japan

WMO: 477150

Lat: 36.182N

Lon: 140.415E

Elev: 117

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	23.2	8.8	8.7	36.9	12.0	10.2	35.3	23.3	43.7	20.7	43.6	2.5	350	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	91.3	79.2	89.2	78.7	86.2	77.5	80.9	87.6	79.6	85.9	78.6	84.3	10.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.6	84.5	77.4	143.3	82.7	76.9	140.8	82.2	44.7	87.9	43.3	86.5	42.1	85.2	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.6	16.3	DB	16.2	94.9	2.1	2.3	14.6	96.5	13.4	97.8	12.2	99.1	10.6	100.7	(4)
(5)			WB	15.0	83.6	2.0	2.2	13.5	85.2	12.4	86.5	11.2	87.8	9.7	89.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	36.7	38.1	44.0	53.4	62.1	68.3	76.0	78.3	71.8	61.5	50.6	41.1	(6)
(7)		DBStd	15.25	4.32	4.95	5.95	6.28	5.25	4.99	5.39	4.57	5.81	5.45	5.65	5.25	(7)
(8)		HDD50	1332	414	335	205	36	0	0	0	0	0	1	61	280	(8)
(9)		HDD65	4089	878	752	650	350	117	23	1	0	10	136	432	740	(9)
(10)		CDD50	3862	0	2	20	138	376	549	806	876	653	358	79	5	(10)
(11)		CDD65	1146	0	0	0	2	28	122	341	411	214	28	0	0	(11)
(12)		CDH74	8304	0	0	0	21	199	605	2620	3458	1293	108	0	0	(12)
(13)		CDH80	2695	0	0	0	1	23	118	928	1266	343	16	0	0	(13)
(14)	Wind	WSAvg	6.8	5.6	6.4	7.5	8.3	7.9	7.0	7.2	7.2	6.4	5.2	5.1	(14)	
(15)	Precipitation	PrecAvg	53.8	2.0	1.9	3.1	4.5	5.5	6.0	5.9	5.2	7.6	6.9	3.3	2.0	(15)
(16)		PrecMax	73.8	4.5	5.9	5.8	8.6	12.2	11.3	11.6	14.8	15.8	21.1	7.6	7.1	(16)
(17)		PrecMin	43.2	0.2	0.3	1.3	1.1	3.0	2.7	1.1	0.9	0.5	0.6	0.0	0.0	(17)
(18)		PrecStd	6.9	1.3	1.3	1.2	2.0	2.0	1.8	2.6	3.3	3.6	4.5	1.8	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	64.1	69.8	76.8	82.7	86.4	93.4	93.6	90.0	82.3	69.9	62.8	(19)
MCWB			48.8	54.3	55.6	61.4	67.7	75.4	79.7	80.2	77.6	70.7	61.0	56.5	(20)	
2%		DB	53.5	55.8	64.5	72.4	78.9	83.8	91.1	91.6	87.6	76.8	66.5	58.8	(21)	
		MCWB	44.1	46.9	53.1	60.3	65.4	72.5	79.0	79.6	77.4	68.6	59.0	50.8	(22)	
5%		DB	50.2	52.2	60.7	69.4	75.5	80.5	88.1	89.6	84.1	73.5	64.4	55.5	(23)	
		MCWB	41.3	43.5	51.5	59.1	64.5	71.1	78.5	79.2	75.8	66.3	57.0	47.4	(24)	
10%		DB	48.0	49.6	57.1	66.0	73.2	77.2	85.8	87.5	80.8	71.3	62.2	52.2	(25)	
		MCWB	39.7	41.5	49.5	56.9	63.6	70.1	77.4	78.8	74.3	64.7	56.3	45.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	56.9	59.7	67.4	72.4	78.4	82.5	83.3	80.1	73.9	64.4	59.1	(27)
MCD B			55.4	61.6	63.5	71.6	76.4	83.5	88.7	89.5	85.9	78.2	67.0	61.6	(28)	
2%		WB	46.4	49.6	57.0	63.4	69.0	75.5	80.6	81.5	78.6	70.7	61.7	52.8	(29)	
		MCD B	50.5	54.2	62.0	68.8	74.4	80.0	87.6	87.9	84.0	74.7	64.6	56.3	(30)	
5%		WB	42.9	45.6	53.8	60.6	67.1	73.5	79.2	80.1	77.2	68.3	59.6	49.3	(31)	
		MCD B	48.0	49.9	59.4	66.1	73.1	77.6	85.7	86.7	82.4	72.0	62.5	53.4	(32)	
10%		WB	40.3	42.4	50.1	58.6	65.2	71.4	78.1	79.0	75.7	65.9	57.3	46.5	(33)	
		MCD B	45.7	47.4	55.1	63.9	70.9	75.8	84.0	84.9	79.9	69.4	60.8	50.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.8	21.0	20.8	20.1	17.5	13.8	13.6	13.9	14.1	16.1	19.5	21.5	(35)
MCDBR			25.2	25.7	26.5	25.1	22.9	19.0	17.9	17.7	17.1	18.4	21.6	23.8	(36)	
MCWBR			17.0	17.8	17.5	14.8	12.4	9.9	8.5	7.9	8.6	11.2	14.9	16.6	(37)	
5% WB		MCDBR	21.7	22.8	23.2	20.2	18.7	15.4	16.4	16.1	15.4	15.1	17.3	20.3	(38)	
		MCWBR	16.6	17.8	18.2	13.9	12.2	9.7	8.8	8.3	8.6	10.6	13.9	16.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.373	0.423	0.495	0.547	0.566	0.600	0.594	0.577	0.508	0.473	0.420	0.377	(40)	
taud		2.247	2.099	1.925	1.852	1.860	1.844	1.908	1.942	2.095	2.115	2.178	2.256	(41)		
Ebn at Noon		253	252	244	239	236	228	228	228	238	233	233	242	(42)		
Edh at Noon		33	43	57	64	65	66	61	58	47	43	36	31	(43)		
(44)	All-Sky Solar Radiation	RadAvg	857	1023	1281	1500	1615	1467	1550	1538	1205	951	781	744	(44)	
(45)		RadStd	50	97	74	138	161	119	248	159	98	91	69	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	+1.03	N/A	N/A	N/A	N/A	N/A	N/A	+252	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page